

System Architecture & Boundaries

StudyMate AI will use a client-server architecture in which the browser handles the user interface while sensitive AI interactions are handled on the server.

5.1 Architecture Overview

User → Next.js Frontend → Application Server → AI Provider → Application Server → Frontend → User

The frontend will provide the interactive user experience, while the server will manage communication with the AI provider and protect sensitive credentials.

5.2 System Boundaries

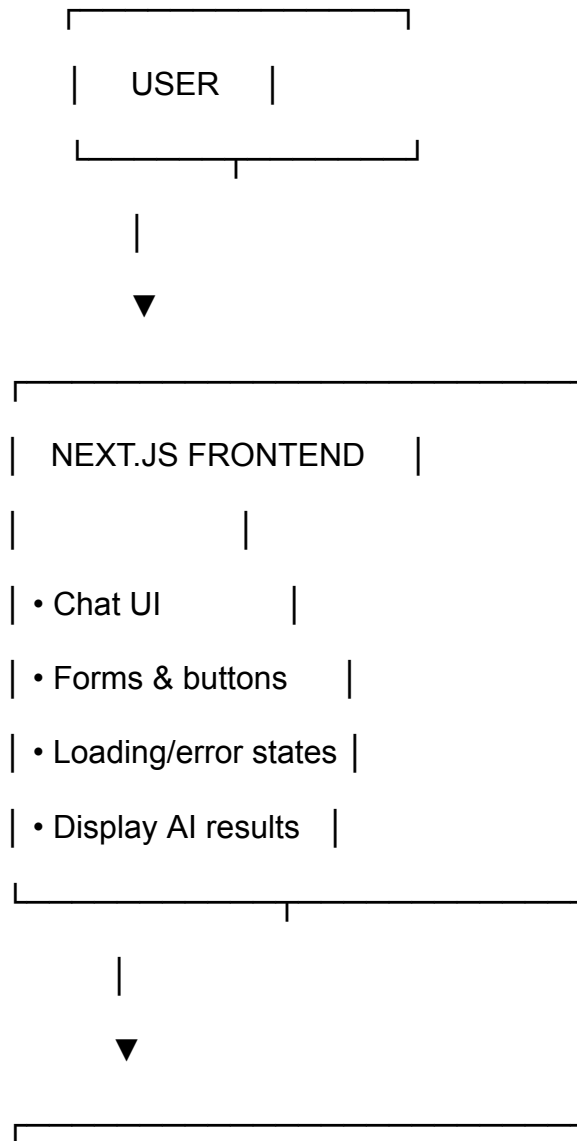
Layer	Responsibilities
Browser / Frontend	User interface, chat interaction, forms, buttons, loading states, displaying AI responses, and client-side input validation.
Application Server	Process requests, validate requests, manage AI provider communication, protect API keys, and control access to server-side resources.
AI Model	Generate explanations, summaries, flashcards, and practice questions based on the user's request and permitted context.
Data Layer	Store approved study materials, conversation/session information, and selected user preferences when persistence is introduced.

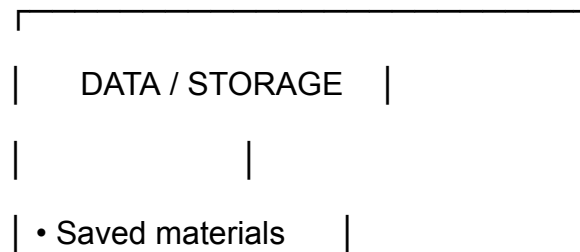
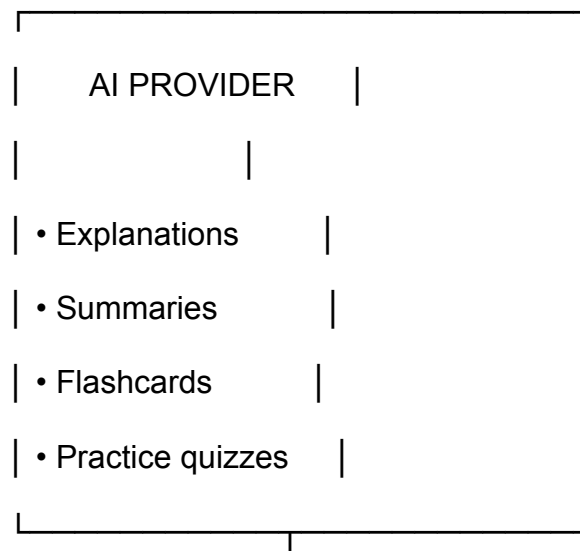
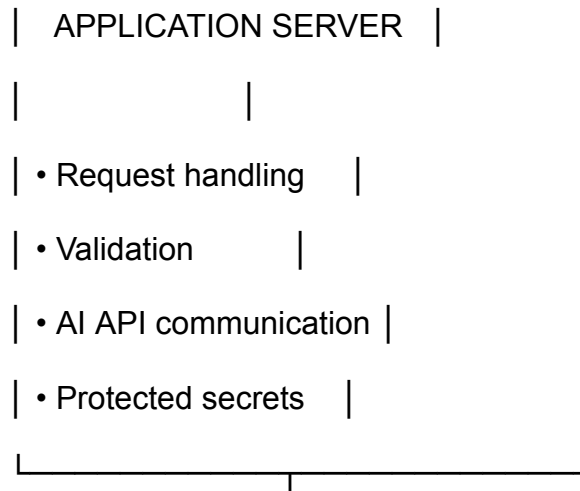
External Services

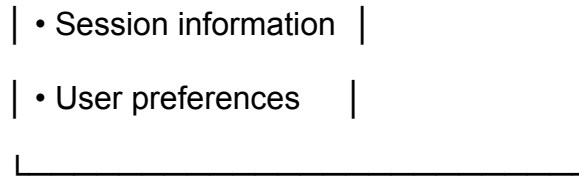
AI provider and deployment infrastructure such as Vercel.

5.3 Security Boundary: Sensitive credentials must remain on the server and must never be exposed directly to the browser. AI provider API keys, database credentials, and other private environment variables will be stored using server-side environment variables. The frontend will communicate with the application's server rather than directly exposing private AI provider credentials.

5.4 Architecture Diagram







Security boundary:

API keys and private credentials remain server-side.

5.5 AI Provider Strategy: The initial implementation will use one selected AI provider, while keeping the AI integration replaceable. The frontend should not be tightly coupled to a specific model provider. The Vercel AI SDK will be considered as the abstraction layer for AI interactions so that the underlying provider can be changed with minimal changes to the frontend application.

5.6 Deployment Boundary: The Next.js application will be connected to GitHub and deployed through Vercel. Future code changes pushed to the repository should trigger an automatic deployment, allowing the live application to remain synchronized with the project's main branch.